Dataset Description

This dataset is used to **detect and classify common lung diseases** from chest X-ray images. The goal is not only to identify **which disease is present**, but also to **locate the affected area in the image** using bounding boxes. Therefore, the task involves both **image classification and object detection**.

The dataset contains **18,000 chest X-ray images** taken in a **postero-anterior (PA) view**. All images are stored in **DICOM format**, which preserves important medical metadata along with the image. To ensure patient privacy, all scans have been **fully de-identified**.

Each image was carefully reviewed and labeled by **multiple experienced radiologists**. This makes the dataset more reliable and realistic, as medical diagnosis often varies slightly between experts. A key challenge of this dataset is handling **multiple annotations from different radiologists** for the same image.

Medical Findings

The images are annotated for **14 critical thoracic (lung and chest) conditions**, including heart enlargement, lung collapse, fluid accumulation, nodules, infections, and fibrosis. If none of these conditions are present, the image is labeled as **"No Finding"**.

The 14 medical conditions include:

- Aortic enlargement
- Atelectasis
- Calcification
- Cardiomegaly
- Consolidation
- Interstitial Lung Disease (ILD)
- Infiltration
- Lung opacity
- Nodule or mass
- Other lesions
- Pleural effusion
- Pleural thickening
- Pneumothorax
- Pulmonary fibrosis

Annotations and Labels

Each abnormal finding in an image is marked with a **bounding box**, which shows the exact location of the disease. Some images may contain **multiple findings**, and each one is recorded separately.

If an image contains **no abnormality**, it is labeled as **"No Finding"**, and a small placeholder bounding box is used to maintain a consistent data format.

Dataset Files

- **train.csv**: Contains detailed annotations for training images, including disease type, bounding box coordinates, and radiologist ID.
- **sample_submission.csv**: Shows the correct format for submitting predictions.

Dataset Columns Explained

- **image_id**: Unique identifier for each X-ray image
- **class_name**: Name of the detected disease or "No finding"
- **class_id**: Numeric ID assigned to each disease class
- **rad_id**: Identifier of the radiologist who made the annotation
- **x_min, y_min**: Top-left corner of the bounding box
- **x_max, y_max**: Bottom-right corner of the bounding box